

Topology and Geometry in Physics – Note 2

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Abstract

This is the *second* note on the course of topology and geometry in physics at the School of Fundamental Physics and Mathematical Sciences, Hangzhou Institute for Advanced Study, UCAS, China. In this note we will learn the differential forms in the first part from Sec. 1 to 3. Particularly, I will also show you how to express the electromagnetism by differential forms in theoretical physics. The second part is the Sec. 4 and 5, we will learn the power of Cartan's exterior covariant differentiation and moving frame method.

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1 Differentiable Manifolds

1.1 Manifolds

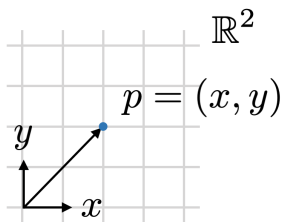


Fig. 1.1: A point p in a Euclidean space $\mathbb{R}^2$.

Euclidean space. We start the discussion on an Euclidean space $\mathbb{R}^2$. Given a point $p \in \mathbb{R}^2$, we can parameterize a point by two parameters x and y as coordinates, i.e., a point p is always written by $p = (x, y) = (x^1, x^2)$. For $\mathbb{R}^n$, a point p can be represented by the coordinates $(x^1, x^2, \dots, x^n)$.

Hausdorff space. A topological space is called a *Hausdorff space* if for any two distinct points p and q , there exists two neighborhoods U and V for $p \in U$ and $q \in V$ such that the intersection of U and V is empty, i.e., $U \cap V = \emptyset$.

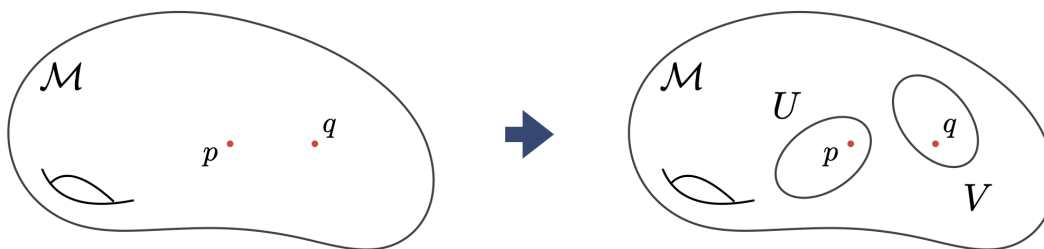


Fig. 1.2: A Hausdorff space $\mathcal{M}$.

Manifold. Let $\mathcal{M}$ be a Hausdorff space. We say that $\mathcal{M}$ is called an n -dimensional *topological manifold* if the neighborhood of any point of $\mathcal{M}$ is isomorphic to an open set in Euclidean space $\mathbb{R}^n$.

Morphism. We will call a map the *morphism*, if it is *structure preserving*.

Local coordinates. Consider a point p in a neighborhood U of a manifold $\mathcal{M}$ with dimension n , we can define a local coordinate system (or a *coordinate chart* or a *coordinate patch* or a *parametrization*) of a neighborhood U of a point p and a *homeomorphism* (an *one-to-one* and *onto* map) φ as a pair (U, φ) . The local coordinates of p is given by a map φ as a set of n -tuple real numbers in a neighborhood $D \subset \mathbb{R}^n$ as

$$\varphi : U \longrightarrow \mathbb{R}^n \tag{1.1a}$$

$$p \longmapsto \varphi(p) := x = (x^1, x^2, \dots, x^n) \in D. \tag{1.1b}$$

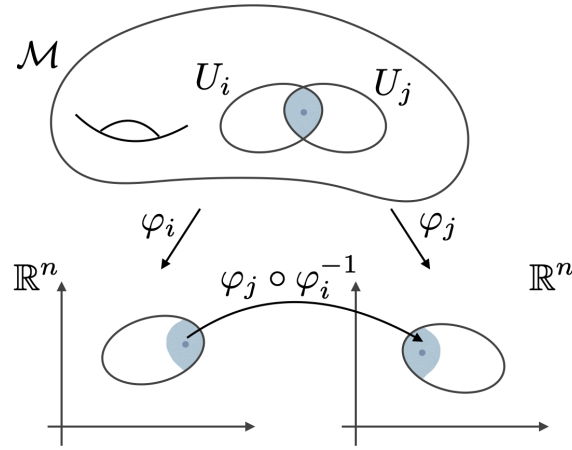


Fig. 1.3: A manifold with coordinate charts.

Coordinate functions. The projection map u^μ sending the coordinate $x = (x^1, x^2, \dots, x^n)$ of a point p to its μ th-component is called the *coordinate function*, i.e., $x^\mu = u^\mu(x) = u^\mu \circ \varphi(p) \equiv \varphi^\mu(p)$.

Coordinate transformation. Consider two local coordinate systems (U_i, φ_i) and (U_j, φ_j) on $\mathcal{M}$. A point p in $U_i \cap U_j$ has two local coordinates $\varphi_i(p) = x_i$ and $\varphi_j(p) = x_j$. We define the *transition function* between i th neighbourhood and j th neighbourhood

$$\begin{cases} \varphi_{ji} := \varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j), \\ \varphi_{ij} := \varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \longrightarrow \varphi_i(U_i \cap U_j), \end{cases} \quad (1.2a)$$

$$\quad (1.2b)$$

and

$$\begin{cases} x_i \longmapsto x_j = \varphi_j \circ \varphi_i^{-1}(x_i) = \varphi_{ji}(x_i), \\ x_j \longmapsto x_i = \varphi_i \circ \varphi_j^{-1}(x_j) = \varphi_{ij}(x_j). \end{cases} \quad (1.3a)$$

$$\quad (1.3b)$$

Example. (A circle S^1 in $\mathbb{R}^2$).

Let S^1 be the circle with radius $r = 1$ in $\mathbb{R}^2$.

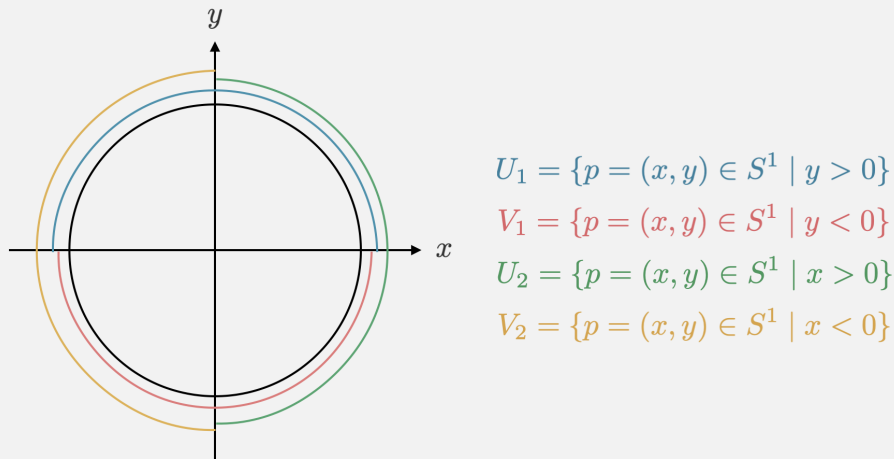


Fig. 1.4: A circle with four charts.

Differentiable manifold. The collection of all local coordinate systems is called an *atlas* of a manifold $\mathcal{M}$ defined by $\mathcal{A} = \{(U_i, \varphi_i)\}$. For any intersection of $U_i \cap U_j$, if φ_{ij} is r times continuously differentiable, then the atlas $\mathcal{A}$ is called the coordinate neighborhood system of class C^r . Therefore, if all the transformation functions are smooth (or differentiable), we say that the atlas $\mathcal{A}$ with the differentiability defines the *differential structure of class C^r* on the topological manifold $\mathcal{M}$.

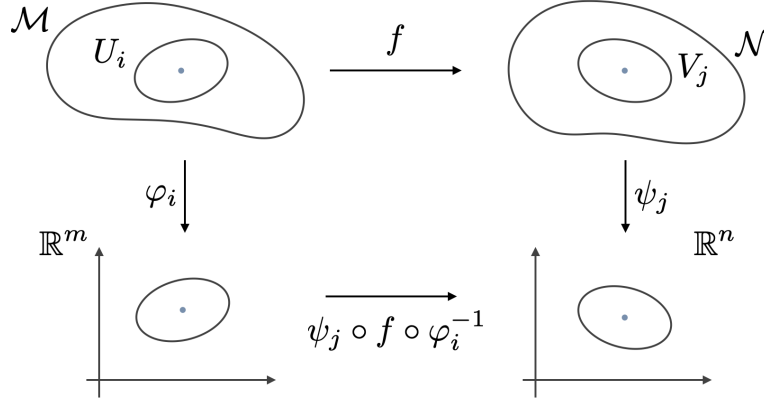


Fig. 1.5: A map from manifold $\mathcal{M}$ to $\mathcal{N}$ with coordinate charts.

Diffeomorphism. Consider another manifold $\mathcal{N}$, the diffeomorphism is a homeomorphism $f : \mathcal{M} \rightarrow \mathcal{N}$ such that f and f^{-1} are both smooth. This is *globally* defined. *Locally*, given two local coordinate systems (U_i, φ_i) and (V_j, ψ_j) on $\mathcal{M}$ and $\mathcal{N}$, respectively, the induced coordinate transformation of the *local* diffeomorphism is read as

$$\begin{cases} x_i \mapsto y_j = \psi_j \circ f \circ \varphi_i^{-1}(x_i) := f_{ji}^{(yx)}(x_i), \\ y_j \mapsto x_i = \varphi_i \circ f^{-1} \circ \psi_j^{-1}(y_j) := f_{ij}^{(xy)}(y_j). \end{cases} \quad (1.4a)$$

$$(1.4b)$$

We note that $f_{ji}^{(yx)} = \psi_j \circ f \circ \varphi_i^{-1}$ and $f_{ij}^{(xy)} = \varphi_i \circ f^{-1} \circ \psi_j^{-1}$ are called the *expressions* of f in the local coordinates φ_i and ψ_j .

1.2 Tangent Vector and Tangent Vector Space

Define a curve $\gamma : I_\varepsilon = (-\varepsilon, \varepsilon) \subset \mathbb{R} \rightarrow \mathcal{M}$, which gives $\tau = \gamma(t)$ on a smooth manifold $\mathcal{M}$. Locally a point is given by $p = \gamma(0)$ on U and the corresponding parameterized coordinates should be

$$\varphi(\gamma(t)) = \varphi \circ \gamma(t) = x(t). \quad (1.5)$$

We assume that $f \in \mathfrak{F}(\mathcal{M}) : \mathcal{M} \rightarrow \mathbb{R}$ is a function on $\mathcal{M}$ where $\mathfrak{F}(\mathcal{M})$ is the algebra of C^k . So the coordinate expression of f on U can be given by

$$f(p) = f(\varphi^{-1}(x)) = f \circ \varphi^{-1}(x) := F(x). \quad (1.6)$$

The maps including the coordinate chart from $\mathbb{R}$ to $\mathcal{M}$ to $\mathbb{R}$ is shown in Fig. 1.6. Combine (1.5) and (1.6), a one-parameter varying function is

$$f(\gamma(t)) = f \circ \gamma(t) = \underbrace{f \circ \varphi^{-1}}_F \circ \underbrace{\varphi \circ \gamma}_x(t) = F \circ x(t) = F(x(t)). \quad (1.7)$$

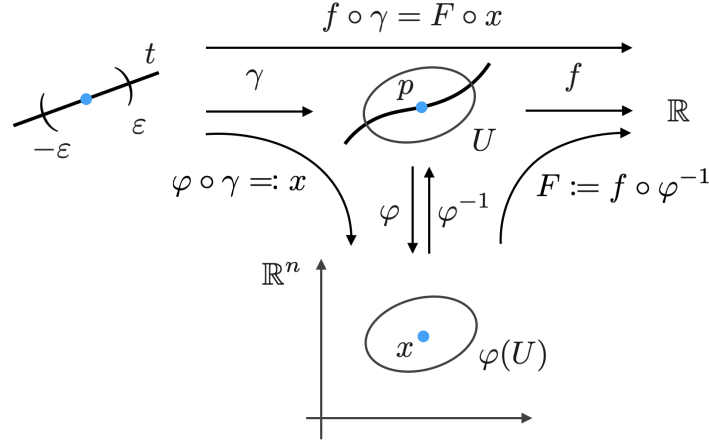


Fig. 1.6: A map of a function f from $\mathbb{R}$ to $\mathbb{R}$.

A *tangent* vector or *velocity* vector filed at point $\gamma(t)$ with the local coordinates $\{x^\mu(t)\}$ in U can be obtained by the so-called derivative map

$$d\gamma : T_t\mathbb{R} \longrightarrow T_{\gamma(t)}\mathcal{M} \quad (1.8)$$

$$\frac{d}{dt} \longmapsto d\gamma\left(\frac{d}{dt}\right) \equiv \frac{d\gamma}{dt} = \dot{\gamma}(t) := X_{\gamma(t)}. \quad (1.9)$$

The tangent vector act on f would be

$$\frac{df(\gamma(0))}{dt} = \frac{dF(x(0))}{dt} = \frac{dx^\mu}{dt}(0) \cdot \frac{\partial F}{\partial x^\mu} \Big|_{\gamma(0)}, \quad (1.10)$$

where $\partial/\partial x^\mu \equiv \partial_\mu$. Hence, a vector field X at $\gamma(0) = p$ is defined as a map which sends a function $f \in \mathfrak{F}(\mathcal{M})$ to $\mathbb{R}$

$$X_p : \mathfrak{F}(\mathcal{M}) \longrightarrow \mathbb{R}, \quad (1.11)$$

and it has an expression of

$$X_p = X_p^\mu (\partial_\mu)_p, \quad (1.12)$$

where the coefficients can be obtained by applying X to the coordinates x^μ , i.e.,

$$\boxed{X_p^\mu = X_p(x^\mu)} = X_p^\nu \frac{\partial x^\mu}{\partial x^\nu} = X_p^\nu \delta_\nu^\mu. \quad (1.13)$$

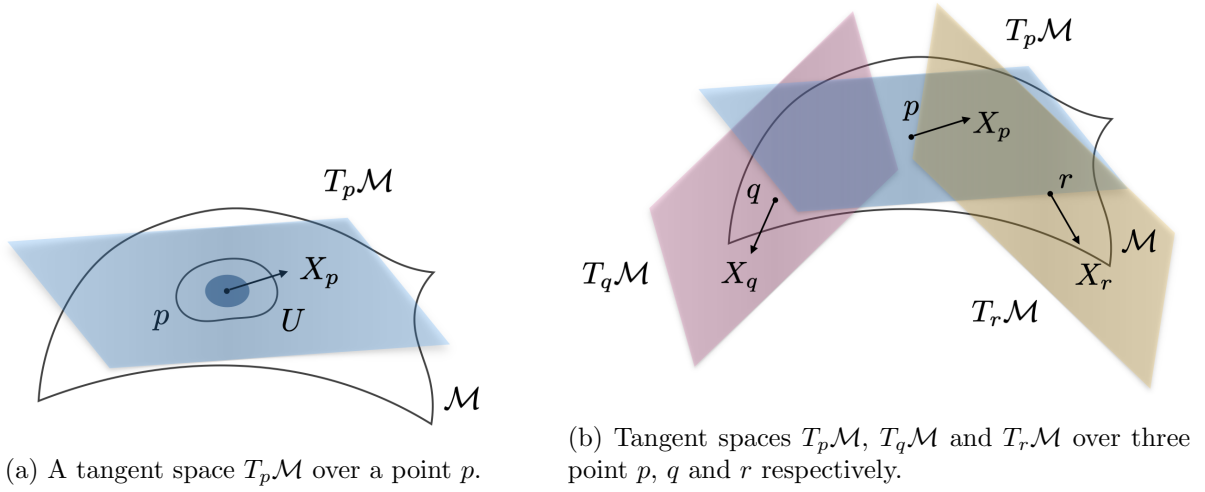


Fig. 1.7: Tangent space.

It can obviously find that $X_p := X_{\gamma(0)}$ is a *derivation* of a smooth manifold $\mathcal{M}$ at point $\gamma(0) = p$, the set of all derivations is denoted by $D_p^r \mathcal{M}$ and

$$X_{\gamma(t)}(f) = X^\mu(t) \partial_\mu F|_{\gamma(t)}. \quad (1.14)$$

Therefore, by identification of $T_{\gamma(t)} \mathcal{M}$ and $D_{\gamma(t)} \mathcal{M}$, components of a tangent vector are given by a set of differential equations

$$X^\mu(t) := \frac{dx^\mu}{dt}(t) \quad (1.15)$$

So the solution x^μ of (1.15) would be obtained

$$\boxed{x^\mu(t) = x^\mu(0) + \int_0^t dt' X^\mu(t')} = x^\mu(0) + t X_p^\mu + \mathcal{O}(t^2), \quad (1.16a)$$

where $X^\mu(t') = X^\mu(0) + t' dX^\mu(t')/dt' + \mathcal{O}(t'^2)$ and $X_p^\mu := X^\mu(0)$. Therefore, the tangent vector of the curve $\gamma(t)$ at p is a directional derivative along the direction of X

$$\dot{\gamma}(0) \equiv X_p = X_p^\mu (\partial_\mu)_p = X^\mu(0) (\partial_\mu)_{\gamma(0)}. \quad (1.17)$$

The discussion above proves that every vector is a linear combination of $\{\partial_\mu\}$. This implies that a vector X represents a differential operator on $\mathcal{M}$. The basis vectors $\{\partial_\mu\}$ given by the local coordinates is called a *natural basis*. Moreover, we can define a so-called *tangent bundle* which is a collection of all tangent spaces

$$T\mathcal{M} = \bigcup_{p \in \mathcal{M}} T_p \mathcal{M}, \quad (1.18)$$

and the set of all *differential* (or *smooth*) vector fields on $\mathcal{M}$ is always denoted by $\mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$.

Consider the bracket operation of vectors X and $Y \in \mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$ acting on $f \in \mathfrak{F}(\mathcal{M})$, then we have

$$[X, Y]f = [X^\mu \partial_\mu, Y^\nu \partial_\nu]f$$

$$\begin{aligned}
&= X^\mu(\partial_\mu Y^\nu)(\partial_\nu f) - Y^\nu(\partial_\nu X^\mu)(\partial_\mu f) \\
&= [X^\mu(\partial_\mu Y^\nu) - Y^\nu(\partial_\nu X^\mu)](\partial_\nu f) \\
&\equiv [X, Y]^\nu(\partial_\nu f),
\end{aligned} \tag{1.19}$$

which shows that $[X, Y]$ is a vector field. It implies that $\mathfrak{X}(\mathcal{M})$ is a Lie algebra. In addition we have the *Jacobi's identity*

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0. \tag{1.20}$$

Differential 1-forms. The map $f : \mathcal{M} \longrightarrow \mathbb{R}$ induces a derived map $(df)_p : T_p\mathcal{M} \longrightarrow T_{f(p)}\mathbb{R} \cong \mathbb{R}$ which is a linear function on a vector

$$(df)_p(aX + bY) = a(df)_p(X) + b(df)_p(Y) \quad \text{for } a, b \in \mathfrak{F}(\mathcal{M}) \text{ and } X, Y \in \mathfrak{X}(\mathcal{M}). \tag{1.21}$$

The function $(df)_p \in T_p^*\mathcal{M}$ is called a *1-form* (a differential form of degree 1). Given a vector field $X \in \mathfrak{X}(\mathcal{M})$, we have

$$(df)_p(X_p) = \frac{df}{dt}(p) = X_p f \in \mathbb{R}. \tag{1.22}$$

Hence, we define a *cotangent bundle* as a set of all cotangent spaces

$$T^*\mathcal{M} = \bigcup_{p \in \mathcal{M}} T_p^*\mathcal{M}. \tag{1.23}$$

The set of all *differential* (or *smooth*) 1-form fields on $\mathcal{M}$ is then denoted by $A^1(\mathcal{M}) = \Gamma(T^*\mathcal{M})$.

Take $f = x^\mu$ and $X_p = (\partial_\nu)_p$. The component of local coordinates is $x^\mu : U \longrightarrow \mathbb{R}$, then $(dx^\mu)_p : T_p U \longrightarrow T_{x^\mu(p)}\mathbb{R} \cong \mathbb{R}$. Consequently we have property

$$(dx^\mu)_p((\partial_\nu)_p) = (\partial_\nu)_p(x^\mu) = \delta_\nu^\mu. \tag{1.24}$$

The equation above implies that $T^*\mathcal{M}$ is the dual space of $T\mathcal{M}$ and $\{dx^\mu\}$ forms a coordinate frame in $T^*\mathcal{M}$. Consequently, we can perform an inner product between $T\mathcal{M}$ and $T^*\mathcal{M}$:

$$\langle \cdot, \cdot \rangle : T\mathcal{M} \times T^*\mathcal{M} \longrightarrow \mathbb{R}, \tag{1.25}$$

$$(X, \omega) \longmapsto \langle X, \omega \rangle. \tag{1.26}$$

Induced maps of vectors and differential 1-forms. Under a *coordinate transformation* between two charts of a point p (1.3), the transformations of natural basis of vectors and differential 1-forms can be understood by using the *chain rule*

$$\left\{ \begin{aligned} \frac{\partial}{\partial x_i^\mu} &= \frac{\partial}{\partial x_j^\nu} \frac{\partial \varphi_{ji}^\nu}{\partial x_i^\mu} \equiv \frac{\partial}{\partial x_j^\nu} \frac{\partial x_j^\nu}{\partial x_i^\mu}, \\ dx_i^\mu &= \frac{\partial \varphi_{ij}^\mu}{\partial x_j^\nu} dx_j^\nu \equiv \frac{\partial x_i^\mu}{\partial x_j^\nu} dx_j^\nu. \end{aligned} \right. \tag{1.27a}$$

$$\left\{ \begin{aligned} dx_i^\mu &= \frac{\partial \varphi_{ij}^\mu}{\partial x_j^\nu} dx_j^\nu \equiv \frac{\partial x_i^\mu}{\partial x_j^\nu} dx_j^\nu. \end{aligned} \right. \tag{1.27b}$$

Given a diffeomorphism between two manifold $f : \mathcal{M} \longrightarrow \mathcal{N}$ such that $p \longmapsto q = f(p)$, a *pushforward* of vectors is induced

$$f_* : T_p\mathcal{M} \longrightarrow T_q\mathcal{N} = T_{f(p)}\mathcal{N}, \tag{1.28a}$$

$$X_p \longmapsto f_* X_p = (f_* X)_q = (f_* X)_{f(p)}. \quad (1.28b)$$

Let us choose coordinate systems in two manifolds (1.4), the pushforward of a basis vector from one to the other is

$$\left. \frac{\partial}{\partial x^\mu} \right|_p \longmapsto f_* \left(\left. \frac{\partial}{\partial x^\mu} \right|_p \right) = \left. \frac{\partial f_{yx}^\nu}{\partial x^\mu} \frac{\partial}{\partial y^\nu} \right|_{q=f(p)} = \left. \frac{\partial f_{yx}^\nu}{\partial x^\mu} \frac{\partial}{\partial y^\nu} \right|_{f(p)} \equiv \left. \frac{\partial y^\nu}{\partial x^\mu} \frac{\partial}{\partial y^\nu} \right|_{f(p)}. \quad (1.29)$$

Similarly, we can define a map with the *backward* direction with respect to the direction of f called a *pullback* of 1-forms induced by the map f

$$f^* : T_q^* \mathcal{N} \longrightarrow T_p^* \mathcal{M} = T_{f^{-1}(q)}^* \mathcal{M}, \quad (1.30a)$$

$$\omega_q \longmapsto f^* \omega_q = (f^* \omega)_p = (f^* \omega)_{f^{-1}(q)}. \quad (1.30b)$$

Therefore the pullback of a coordinate 1-form is

$$dy^\mu|_q \longmapsto f^*(dy^\mu|_q) = \left. \frac{\partial f_{yx}^\mu}{\partial x^\nu} dx^\nu \right|_{p=f^{-1}(q)} = \left. \frac{\partial f_{yx}^\mu}{\partial x^\nu} dx^\nu \right|_{f^{-1}(q)} \equiv \left. \frac{\partial y^\mu}{\partial x^\nu} dx^\nu \right|_{f^{-1}(q)}. \quad (1.31)$$

As discussions above, we compare properties of diffeomorphism, pushforward and pullback together as follows

$$\text{Diffeomorphism } f : \mathcal{M} \text{ (left)} \longrightarrow \mathcal{N} \text{ (right)}, \quad (1.32a)$$

$$\text{Pushforward } f_* : T_p \mathcal{M} \text{ (left)} \longrightarrow T_p \mathcal{N} \text{ (right)}, \quad (1.32b)$$

$$\text{Pullback } f^* : T_p^* \mathcal{N} \text{ (right)} \longrightarrow T_p^* \mathcal{M} \text{ (left)}. \quad (1.32c)$$

It is apparent that the direction of pushforward f_* is the *same* as diffeomorphism f , and the direction of pullback f^* , however, is the *inversed*. That is why we called the objects with subscript the *covariant* fields and with superscript the *contravariant* fields.

Tensor fields. Suppose V is a n -dimensional vector space over $\mathfrak{F}$ and its dual space is V^* . We define (r, s) -type tensor fields V_s^r (or $V^{(r,s)}$) by the so-called tensor product

$$V_s^r : \underbrace{V^* \times \cdots \times V^*}_{r \text{ terms}} \times \underbrace{V \times \cdots \times V}_{s \text{ terms}} \longrightarrow \mathbb{R}. \quad (1.33)$$

Take e_μ as the basis of V and ϑ^μ as the basis of V^* . A tensor field $T \in V_s^r$ is then given by

$$K = K^{\mu_1 \cdots \mu_r}_{\nu_1 \cdots \nu_s} e_{\mu_1} \otimes \cdots \otimes e_{\mu_r} \otimes \vartheta^{\nu_1} \otimes \cdots \otimes \vartheta^{\nu_s}, \quad (1.34)$$

where

$$K^{\mu_1 \cdots \mu_r}_{\nu_1 \cdots \nu_s} = K(\vartheta^{\mu_1}, \dots, \vartheta^{\mu_r}, e_{\nu_1}, \dots, e_{\nu_s}). \quad (1.35)$$

Obviously we have $V_0^0 = \mathfrak{F}$, $V_0^1 = V$ and $V_1^0 = V^*$.

Multilinear algebra. The spaces $\Lambda^r V$ and $\Lambda^r V^*$ being subspaces of V_0^r and V_r^0 are the set of all alternating contravariant and covariant tensor fields, respectively. Particularly, the space of all vectors is identified by $\Lambda^1 V = V = V_0^1$, while the space of all 1-forms is identified by $\Lambda^1 V^* = V^* = V_1^0$.

1.3 1-Parameter Group of Transformations

A 1-parameter group. Consider a mapping given by

$$\sigma : \mathbb{R} \times \mathcal{M} \longrightarrow \mathcal{M}, \quad (1.36)$$

$$(t, p) \longmapsto \sigma(t, p). \quad (1.37)$$

we can also have $\mathcal{M}$ by $\varphi_t(p) \equiv \gamma_p(t) := \sigma(t, p)$ where

$$\varphi_t : \mathcal{M} \ni p \longmapsto q = \varphi_t(p) \in \mathcal{M}, \quad (1.38a)$$

$$\gamma_p : \mathbb{R} \ni t \longmapsto \gamma_p(t) \in \mathcal{M}. \quad (1.38b)$$

For any $t \in \mathbb{R}$, the map φ_t is called the 1-parameter group of (differentiable) transformations of $\mathcal{M}$, which satisfies the following conditions:

- $\varphi_0 = \text{id}_{\mathcal{M}}$, i.e., $\varphi_0(p) = \sigma(0, p) = p$;
- $\varphi_t \circ \varphi_s = \varphi_{t+s}$ for any $t, s \in \mathbb{R}$.

The second condition can be realized as follows:

$$(\varphi_t \circ \varphi_s)(p) = \gamma_{\varphi_s(p)}(t) = \gamma_p(t + s) = \sigma(t + s, p) = \varphi_{t+s}(p). \quad (1.39)$$

It can be shown that the conditions above implies $\varphi_{-t} = \varphi_t^{-1}$ by substituting $s = -t$. Therefore, $\varphi_t : \mathcal{M} \longrightarrow \mathcal{M}$ is a *diffeomorphism* from $\mathcal{M}$ to itself, which forms a diffeomorphism group denoted by $\text{Diff}(\mathcal{M})$. This diffeomorphism also induces a map of tangent vectors $d\varphi_t : T\mathcal{M} \longrightarrow T\mathcal{M}$:

$$d\varphi_t(X_p) = X_q = X_{\varphi_t(p)}. \quad (1.40)$$

Let X be a vector field on $\mathcal{M}$, a parameterized curve γ_p called the *orbit* of φ_t through $p \in \mathcal{M}$. By writing as $\varphi_t(p)$, it can also be called an *integral curve* (or a *flow*) of X through a point p at $t = 0$. Therefore, we say that the transformation induces a *complete* vector field X on $\mathcal{M}$. Conversely, any $X \in \mathfrak{X}(\mathcal{M})$ generates a 1-parameter group of transformations of $\mathcal{M}$, $\varphi_t \in \text{Diff}(\mathcal{M}) : \mathcal{M} \longrightarrow \mathcal{M}$.

A *local* version of transformation can be defined in a same way, if t is defined in a neighborhood of 0 and p is in an open set U of $\mathcal{M}$. The induced vector field is

$$X(t, p) \equiv X_{\gamma_p(t)} = d\gamma_p \left(\frac{d}{dt} \right) = \frac{d\gamma_p(t)}{dt} = \frac{d}{dt} \sigma(t, p). \quad (1.41)$$

Let x be a local coordinates of U , then the vector component in U should be

$$X^\mu(t, p) = X_{\gamma_p(t)}(x^\mu) = d\gamma_p \left(\frac{d}{dt} \right) (x^\mu) = \frac{d}{dt} (x^\mu \circ \gamma_p(t)) = \frac{d}{dt} x^\mu(\sigma(t, p)). \quad (1.42)$$

Similarly, any $X \in \mathfrak{X}(U)$ generates a local 1-parameter group of *local* transformations of U , $\varphi_t \in \text{Diff}(U) : U \longrightarrow U$. By Taylor expansion, we have a local diffeomorphism from a point p to $\sigma(t, p) \equiv q$:

$$q = \sigma(t, p) = \sigma(0, p) + t X(0, p) + \mathcal{O}(t^2) = p + t X_p + \mathcal{O}(t^2), \quad (1.43)$$

which can be expressed in the local coordinate system in linear term

$$x^\mu(t, p) = x^\mu(0, p) + t X^\mu(0, p) \quad \text{or} \quad x^\mu(q) = x^\mu(p) + t X_p^\mu, \quad (1.44)$$

where we have defined $x^\mu(t, p) := x^\mu(\sigma(t, p)) \equiv x^\mu(q)$.

Exponential map. The group...

1.4 Lie Derivative

Infinitesimal transformation. Infinitesimally we have the exponential map from the Lie algebra to its Lie group

$$\varphi_t = \exp(t X) = 1 + t X + \frac{1}{2!} t^2 X^2 + \dots \quad (1.45)$$

This infinitesimal transformation p to

$$\varphi_t^\mu(p) \approx x^\mu + t X(x^\mu) = x^\mu + t X^\nu \underbrace{\partial_\nu x^\mu}_{\delta_\nu^\mu} = x^\mu + t \xi^\mu \quad (1.46)$$

Lie derivative for a scalar. The Lie derivative for a scalar is defined by

$$\boxed{\mathcal{L}_X f = \lim_{t \rightarrow 0} \frac{(\varphi_t)^* f - f}{t} = \left. \frac{d}{dt} (\varphi_t)^* f \right|_{t=0}} \quad (1.47)$$

Act it on a point p , we get

$$(\mathcal{L}_X f)(p) = \lim_{t \rightarrow 0} \frac{((\varphi_t)^* f)(p) - f(p)}{t} = \lim_{t \rightarrow 0} \frac{(f \circ \varphi_t)(p) - f(p)}{t} \quad (1.48)$$

where

$$(f \circ \varphi_t)(p) = \underbrace{f(\varphi_t(p))}_{f(q)} = f(p + t X_p) = f(p) + t X_p(f). \quad (1.49)$$

So the equation above becomes

$$(\mathcal{L}_X f)(p) = \lim_{t \rightarrow 0} \frac{f(p) + t X_p(f) - f(p)}{t} = X_p(f) = X f(p) \quad (1.50)$$

Finally the Lie derivative for a scalar is

$$\boxed{\mathcal{L}_X f = X f.} \quad (1.51)$$

Lie derivative for a vector. The Lie derivative for a vector is defined by

$$\boxed{\begin{aligned} \mathcal{L}_X Y &= \lim_{t \rightarrow 0} \frac{Y - (\varphi_t)_* Y}{t} \\ &= \lim_{t \rightarrow 0} (\varphi_t)_* \frac{(\varphi_t^{-1})^* Y - Y}{t} = \lim_{t \rightarrow 0} \frac{(\varphi_{-t})^* Y - Y}{t} = \left. \frac{d}{dt} (\varphi_{-t})^* Y \right|_{t=0}, \end{aligned}} \quad (1.52)$$

where we have use the fact of $(\varphi_0)_* = \text{id}$. Act this on a function f more precisely leads to

$$(\mathcal{L}_X Y)_q f = \lim_{t \rightarrow 0} \frac{Y_q f - ((\varphi_t)_* Y)_q f}{t} \quad (1.53)$$

We note that the second term can be realized by

$$((\varphi_t)_* Y)_q f = (Y(f \circ \varphi_t))(q(t)) = (Yf)(q(t)) + t \cdot Y(Xf)(q(t)) \quad (1.54)$$

where we have used $q(t) := \varphi_t^{-1}(q) \equiv p$ and

$$f \circ \varphi_t = f + t \cdot (Xf). \quad (1.55)$$

Hence, we have

$$\begin{aligned} (\mathcal{L}_X Y)_q f &= \lim_{t \rightarrow 0} \frac{(Yf)(q) - [(Yf)(q(t)) + t \cdot Y(Xf)(q(t))]}{t} \\ &= \lim_{t \rightarrow 0} \underbrace{\frac{(Yf)(q) - (Yf)(q(t))}{t}}_{\text{Let } g = Yf, \lim_{t \rightarrow 0} \frac{g(q) - g(q(t))}{t} = X_{q(0)}(g) = X_q(g)} - Y(Xf)(q(0)) \\ &= X(Yf)(q) - Y(Xf)(q) \\ &= [X, Y]_q f. \end{aligned} \quad (1.56)$$

As a result, the Lie derivative for a vector is

$$\boxed{\mathcal{L}_X Y = [X, Y].} \quad (1.57)$$

Lie derivative for a 1-form.

$$\boxed{\mathcal{L}_X \omega = \lim_{t \rightarrow 0} \frac{(\varphi_t)^* \omega - \omega}{t}} \quad (1.58)$$

Lie derivative for a tensor. For tensor fields, we have a operation denoted by $(\tilde{\varphi}_t)^*$ which combines operations of $(\varphi_{-t})_*$ for vectors and $(\varphi_t)^*$ for scalars and 1-forms

$$\boxed{\mathcal{L}_X T = \lim_{t \rightarrow 0} \frac{(\tilde{\varphi}_t)^* T - T}{t}} \quad (1.59)$$

Lie differential of a tensor. Finally we can define the *Lie differential* of a tensor T

$$\delta_L T = (\tilde{\varphi}_t)^* T - T \quad \text{or} \quad T - (\tilde{\varphi}_t)_* T, \quad (1.60)$$

then we can conclude that

$$\boxed{\delta_L T = \delta t \mathcal{L}_X T \quad \text{or} \quad \mathcal{L}_X T = \frac{\delta_L T}{\delta t}} \quad (1.61)$$

The details of calculation are associated with the *infinitesimal point transformation* which can be found in Appendix ??.

1.5 Dual Vector Space

Let f be a function over a field $\mathfrak{F}$. A field $\mathfrak{F}$ can be either $\mathbb{R}$ or $\mathbb{C}$. We call f is *linear* if

$$f(ax + by) = af(x) + bf(y), \quad a, b \in \mathfrak{F}. \quad (1.62)$$

Let V be a n -dimensional vector space and a map $\vec{f}^* : V \longrightarrow \mathfrak{F}$ is a $\mathfrak{F}$ -valued *function* on V . For $\vec{v}_1, \vec{v}_2 \in V$ and $a_1, a_2 \in \mathfrak{F}$ such that

$$\vec{f}^*(a_1\vec{v}_1 + a_2\vec{v}_2) = a_1\vec{f}^*(\vec{v}_1) + a_2\vec{f}^*(\vec{v}_2), \quad (1.63)$$

we call $\vec{f}^*$ is a $\mathfrak{F}$ -valued linear function on V . The collection of all $\vec{f}^*$ s on V forms a n -dimensional vector space, which is called *dual vector space* of V , denoted by V^* . Let $\{\vec{e}_i\}$ be a set of basis vector of V , a vector can be written by

$$\vec{v} = \vec{e}_i v_i. \quad (1.64)$$

Let $\vec{f}^* \in V^*$ such that

$$\vec{f}^*(v) = \vec{f}^*(\vec{e}_i v_i) = \vec{f}^*(\vec{e}_i) v_i. \quad (1.65)$$

Define a set $\{\vec{e}_i^*\}$ to be *linear functions* of V such that

$$\vec{e}_i^*(\vec{e}_j) = \delta_{ij} \quad (1.66a)$$

$$\equiv \langle \vec{e}_i^*, \vec{e}_j \rangle. \quad (1.66b)$$

which we called the *dual relation*, and

$$\vec{e}_i^*(v) = v_i. \quad (1.67)$$

We can define

$$f_i^* := \vec{f}^*(\vec{e}_i) \quad (1.68)$$

Hence, we have

$$\vec{f}^*(v) = f_i^* v_i = f_i^* \vec{e}_i^*(v) \implies \boxed{\vec{f}^* = f_i^* \vec{e}_i^*}. \quad (1.69)$$

Consider that the differentiation on the both sides of $\vec{e}_i^*(\vec{e}_j) = \delta_{ij}$:

$$\begin{aligned} 0 &= \mathbf{d}(\vec{e}_i^*(\vec{e}_j)) = (\mathbf{d}\vec{e}_i^*)\vec{e}_j + \vec{e}_i^*(\mathbf{d}\vec{e}_j) \\ &= \Gamma^{*k}_i(\vec{e}_k^*(\vec{e}_j)) + \Gamma^k_j(\vec{e}_i^*(\vec{e}_k)) = \Gamma^{*j}_i + \Gamma^i_j. \end{aligned} \quad (1.70)$$

Hence, the connection (form) in the dual space is

$$\boxed{\Gamma^{*j}_i = -\Gamma^i_j}. \quad (1.71)$$

^{†1} *Quantum mechanically* a vector and a dual vector introduced in a *Hilbert space* $\mathcal{H}$ are represented by a *bra* vector and a *ket* vector by the *Dirac's notation*, respectively:

$$\boxed{\vec{e}_i^* \longrightarrow \langle i|} \quad \text{and} \quad \boxed{\vec{e}_j \longrightarrow |j\rangle}, \quad \text{which leads to} \quad \boxed{\langle \vec{e}_i^*, \vec{e}_j \rangle \longrightarrow \langle i|j \rangle = \delta_{ij}}.$$

However, in addition to the transpose operation T , quantum mechanical bra vectors further need the *complex conjugate* operation $*$, i.e., $\langle i| = |i\rangle^\dagger := (|i\rangle^\mathsf{T})^*$ where $\dagger$ is called the *Hermitian conjugate* operation.

New notation. In order to unify the notation, we use $\{e_i\}$ without an arrow as the basis of V and $\{\vartheta^i\}$ as the dual basis of V^* instead of $\{\vec{e}_i^*\}$, while, for the notation of components, the upper and lower indices are, respectively, used for components of a vector and a dual vector. That means we have the following replacement:

$$\begin{cases} \vec{e}_i & \longrightarrow & e_i & \text{ and } & v_i & \longrightarrow & v^i, \end{cases} \quad (1.72a)$$

$$\begin{cases} \vec{e}_i^* & \longrightarrow & \vartheta^i & \text{ and } & w_i^* & \longrightarrow & w_i. \end{cases} \quad (1.72b)$$

Hence, the new notation of a vector $\vec{v}$ and a dual vector $\vec{w}^*$ is written, respectively, by

$$\boxed{\vec{v} = e_i v^i \in V} \quad \text{and} \quad \boxed{\vec{w}^* = w_i \vartheta^i \in V^*} \quad (1.73)$$

and the dual relation (1.66) becomes

$$\boxed{\vartheta^i(e_j) = \delta_j^i \equiv \langle \vartheta^i, e_j \rangle.} \quad (1.74)$$

Furthermore, we have

$$d\vec{e}_i^* = \vec{e}_k^* \Gamma^{*k}_i \xrightarrow{\text{T}} d\vartheta^i = \Gamma^{*i}_k \vartheta^k \implies \boxed{(\text{old})\Gamma^{*k}_i \xrightarrow{\text{T}} (\text{new})\Gamma^{*i}_k = \left((\text{old})\Gamma^{*k}_i\right)^\text{T}}, \quad (1.75)$$

or, more precisely, in the matrix form

$$\underbrace{\begin{pmatrix} d\vec{e}_1^* & d\vec{e}_2^* & \cdots & d\vec{e}_i^* & \cdots & d\vec{e}_n^* \end{pmatrix}}_{(d\mathbf{e}_i^*)} = \underbrace{\begin{pmatrix} \vec{e}_1^* & \vec{e}_2^* & \cdots & \vec{e}_k^* & \cdots & \vec{e}_n^* \end{pmatrix}}_{(\mathbf{e}_k^*)} \underbrace{\begin{pmatrix} \Gamma^{*1}_1 & \Gamma^{*1}_2 & \cdots & \Gamma^{*1}_i & \cdots & \Gamma^{*1}_n \\ \Gamma^{*2}_1 & \Gamma^{*2}_2 & \cdots & \Gamma^{*2}_i & \cdots & \Gamma^{*2}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*k}_1 & \Gamma^{*k}_2 & \cdots & \Gamma^{*k}_i & \cdots & \Gamma^{*k}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*n}_1 & \Gamma^{*n}_2 & \cdots & \Gamma^{*n}_i & \cdots & \Gamma^{*n}_n \end{pmatrix}}_{(\Gamma^{*k}_i)} \quad (1.76a)$$

$$\xrightarrow{\text{T}} \underbrace{\begin{pmatrix} d\vartheta^1 \\ d\vartheta^2 \\ \vdots \\ d\vartheta^i \\ \vdots \\ d\vartheta^n \end{pmatrix}}_{(d\boldsymbol{\theta}^i)} = \underbrace{\begin{pmatrix} \Gamma^{*1}_1 & \Gamma^{*1}_2 & \cdots & \Gamma^{*1}_k & \cdots & \Gamma^{*1}_n \\ \Gamma^{*2}_1 & \Gamma^{*2}_2 & \cdots & \Gamma^{*2}_k & \cdots & \Gamma^{*2}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*i}_1 & \Gamma^{*i}_2 & \cdots & \Gamma^{*i}_k & \cdots & \Gamma^{*i}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*n}_1 & \Gamma^{*n}_2 & \cdots & \Gamma^{*n}_k & \cdots & \Gamma^{*n}_n \end{pmatrix}}_{(\Gamma^{*i}_k)} \underbrace{\begin{pmatrix} \vartheta^1 \\ \vartheta^2 \\ \vdots \\ \vartheta^k \\ \vdots \\ \vartheta^n \end{pmatrix}}_{(\boldsymbol{\theta}^k)}. \quad (1.76b)$$

It can be check that the connection form in the new notation in the dual vector space is has the same result in terms of the connection form in the vector space. Applying the differentiation on (1.74) leads to

$$0 = d\vartheta^i(e_j) = (d\vartheta^i)e_j + \vartheta^i(de_j)$$

$$= \Gamma^{*i}_k \vartheta^k(e_j) + \Gamma^{ki}_{j\vartheta}(e_k) = \Gamma^{*i}_j + \Gamma^i_j \quad \Longrightarrow \quad \boxed{\Gamma^{*i}_j = -\Gamma^i_j}. \quad (1.77)$$

More precisely, it is read by

$$\boxed{^{(\text{new})}\Gamma^{*i}_j = ^{(\text{ond})}\Gamma^{*j}_i = -\Gamma^i_j}, \quad (1.78)$$

which is consist with (1.71).

In physics, we always write vectors and dual vectors by their components only, a vector component v^i carries *upper* index called a *contravariant vector*, while a dual vector component w_i carries *lower* index called a *covariant vector* (or *covector* simply):

$$\text{A contravariant vector: } \vec{v} \longrightarrow v^i \equiv \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix}; \quad (1.79a)$$

$$\text{A covariant vector: } \vec{w}^* \longrightarrow w_i \equiv (w_1 \ w_2 \ \cdots \ w_n). \quad (1.79b)$$

Alternatively, the following realization of vectors and forms is given. The upper index can be regarded as the index of a column, while the lower index represents a row index, i.e., vectors and covectors are respectively realized by the product of a row matrix and a column matrix

$$\vec{v} = e_i v^i \equiv (e_1 \ e_2 \ \cdots \ e_n) \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix} \quad \text{and} \quad \vec{w}^* = w_i \vartheta^i \equiv (w_1 \ w_2 \ \cdots \ w_n) \begin{pmatrix} \vartheta^1 \\ \vartheta^2 \\ \vdots \\ \vartheta^n \end{pmatrix}. \quad (1.80)$$

By definition, a covector is a function of a vector. One can also take the covector as an operator acting on a vector realized by

$$\begin{aligned} \vec{w}^*(\vec{v}) &= w_i \vartheta^i(e_j v^j) = w_i \vartheta^i(e_j) v^j = w_i \delta^i_j v^j = w_j v^j \\ &= w_1 v^1 + w_2 v^2 + \cdots + w_n v^n \equiv (w_1 \ w_2 \ \cdots \ w_n) \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix}. \end{aligned} \quad (1.81)$$

Tensor.

1.6 Multi-linear Space

A map $f^* : V \longrightarrow \mathfrak{F}$

Exterior (or wedge) product. Let V be a n -dimensional vector space, A space of tensor can be constructed by the tensor product

$$V^2 = V \otimes V \quad (1.82)$$

The exterior product

$$\Lambda^2 V := V^2 \quad (1.83)$$

$$V^r = \underbrace{V \otimes V \otimes \cdots V}_{r \text{ times}} := \otimes^r V \quad (1.84)$$

1.7 Cartan's Lemma

Now we would like to introduce a very useful lemma of Cartan.

Lemma (Cartan's lemma). Consider a set of linearly independent frame $\{\vec{e}_i\}$ (or coframe $\{\vartheta^i\}$) with $i = 1, \dots, p$ ($p < n$) in n -dimensional space $\mathcal{M}$ and $\{\vec{E}_i\}$ is another set of frame. If $\vec{e}^i \wedge \vec{E}_i = \vec{e}^1 \wedge \vec{E}_1 + \cdots + \vec{e}^p \wedge \vec{E}_p = 0$, then $\vec{E}_i = c_{ij} \vec{e}^j$ and $c_{ij} = c_{ji}$.

Proof. We set the linearly independent frame in $\mathcal{M}$ by extending to n -tuple from $\vec{e}_i$ given by

$$\underbrace{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_p}_p, \underbrace{\vec{e}_{p+1}, \dots, \vec{e}_n}_{n-p} \quad (1.85)$$

with index α labeled components $p+1, p+2, \dots, n$. We assume that $\vec{E}_i$ is expanded by frame in $\mathcal{M}$ as

$$\vec{E}_i = c_{ij} \vec{e}^j + c_{i\alpha} \vec{e}^\alpha. \quad (1.86)$$

According to $\vec{e}^i \wedge \vec{E}_i = 0$, we have

$$\begin{aligned} 0 &= \vec{e}^i \wedge \vec{E}_i = c_{ij} \vec{e}^i \wedge \vec{e}^j + c_{i\alpha} \vec{e}^i \wedge \vec{e}^\alpha \\ &= \frac{1}{2} \underbrace{(c_{ij} - c_{ji})}_0 \vec{e}^i \wedge \vec{e}^j + \underbrace{c_{i\alpha}}_0 \vec{e}^i \wedge \vec{e}^\alpha. \end{aligned} \quad (1.87)$$

Therefore, we obtain the coefficients of $\vec{E}_i$

$$\begin{cases} c_{ij} = c_{ji}, & (1.88a) \\ c_{i\alpha} = 0, & (1.88b) \end{cases}$$

which means that $\vec{E}_i = c_{ij} \vec{e}^j$ is constructed by $\{\vec{e}_j\}$ only. $\square$